Connection Between Longitudinal Magnetic Permeability and the Vector Potential of Total Current and Fermi's Chronometric Field: From Contradictions of Classical Electrodynamics to Modified Theory

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March 7, 2026

Abstract

This paper presents a detailed investigation of the connection between the longitudinal magnetic permeability $\mu_{\parallel}$ introduced in Nikolaev's electrodynamics and two fundamental concepts: (1) the vector potential of total current (conduction + displacement) and (2) Fermi's chronometric correction to the action integration measure. Based on analysis of the author's series of works (2023–2025), it is shown that the problem of "gauge juggling" in the classical derivation of the d'Alembert wave equation can be solved through introduction of $\mu_{\parallel}$, which is physically interpreted as a manifestation of the world tube geometry of an accelerating charge in Fermi's formalism (1923). An explicit expression for $\mu_{\parallel}$ is derived through Fermi's type B variation, equivalence of Nikolaev's and Fermi's approaches is demonstrated, and applications to ponderomotive force calculations in MenDrive-type resonators and electromagnetic impulses of centrally-symmetric plasma explosions are discussed.

1 Introduction

Classical electrodynamics, despite its century-long history, continues to contain a number of fundamental contradictions that become particularly evident when considering problems with accelerating charges and non-stationary fields. Key problems include:

1. **The problem of "gauge juggling"** in the derivation of the d'Alembert wave equation [5];
2. **The 4/3 problem** of electromagnetic mass [1];
3. **Discrepancy of electromagnetic impulse sign** in experiments with centrally-symmetric plasma explosions [9];
4. **Impossibility of obtaining thrust** in closed resonators within standard theory [7].

In a series of the author's works (2023–2025), it was shown that all these problems can be solved through introduction of **longitudinal magnetic permeability** $\mu_{\parallel}$, which

differs from the ordinary transverse permeability μ. The goal of this paper is to establish a strict mathematical connection between $\mu_{\|}$ and two independent concepts: the vector potential of total current and Fermi's chronometric field.

2 Contradictions in the Classical Derivation of the Wave Equation

2.1 Analysis of Tamm's Derivation

In work [5], a detailed analysis of the wave equation derivation in I.E. Tamm's textbook "Fundamentals of the Theory of Electricity" is conducted. It is shown that when deriving displacement current, the electrostatic relation is used:

$$\mathrm{div}\vec{D} = 4\pi\rho, \quad \vec{E} = -\nabla\phi \tag{1}$$

which corresponds to **Coulomb gauge** ($\mathrm{div}\vec{A} = 0$). However, when transitioning to the wave equation, the full expression for the field is used:

$$\vec{E} = -\nabla\phi - \frac{1}{c}\frac{\partial\vec{A}}{\partial t} \tag{2}$$

and the Lorentz gauge condition:

$$\mathrm{div}\vec{A} = -\frac{1}{c}\frac{\partial\phi}{\partial t} \tag{3}$$

Contradiction: Within a single derivation, there is an implicit transition between different definitions of time and space. As noted in [5]:

> "*When deriving displacement current, the electrostatic relation is used (Coulomb gauge), and when transitioning to the wave equation — the full expression for the field is used (Lorentz gauge). Classical electrodynamics 'patches' this hole with mathematical manipulations to obtain the correct wave equation, but the physical justification of the transition becomes blurred.*"

2.2 Problem of Retarded Potentials

In work [3], Tamm's proof (§96) that retarded potentials satisfy the Lorentz condition is analyzed. Two mathematical incorrectnesses are identified:

1. **Incorrect extension of integration** to all infinite space with nullification of the surface integral:

$$\int \mathrm{div}'_q \frac{\vec{j}(t')}{R} dV' = \oint \frac{j_n(t')}{R} dS' \longrightarrow 0 \tag{4}$$

   This is physically incorrect because displacement currents occupy all infinite space, but in the classical derivation they are not accounted for during integration.

2. **Incorrect equating** $\frac{\partial t'}{\partial t} = 1$. For a moving charge, the correct expression is:

$$\frac{\partial t'}{\partial t} = \frac{1}{1 - \frac{\vec{v}\cdot\vec{R}}{Rc}} = \frac{R}{R^*} \tag{5}$$

   where $R^* = R - \frac{\vec{v}\cdot\vec{R}}{c}$ — the Liénard-Wiechert radius.

3 Vector Potential of Total Current

3.1 Contradiction in Coulomb Gauge

In work [4], it is shown that for the classical vector potential created only by conduction current:

$$\vec{A}_{\text{cond}} = \frac{1}{c}\int \frac{\vec{j}_{\text{cond}}(t')}{R} dV' \tag{6}$$

the Coulomb gauge condition ($\text{div}\vec{A} = 0$) is not satisfied:

$$\text{div}\vec{A}_{\text{cond}} \neq 0 \tag{7}$$

Solution: The **total vector potential** created by total current (conduction + displacement) is introduced:

$$\vec{A}_{\text{total}} = \frac{1}{c}\int \frac{(\vec{j}_{\text{cond}} + \vec{j}_{\text{disp}})}{R} dV' \tag{8}$$

where displacement current:

$$\vec{j}_{\text{disp}} = \frac{1}{4\pi}\frac{\partial \vec{E}}{\partial t} \tag{9}$$

For total current, $\text{div}\vec{j}_{\text{total}} = 0$ holds, and Coulomb gauge becomes applicable **only to the total potential**.

3.2 Connection with Longitudinal Permeability

In work [6], a modified fourth Maxwell equation with Nikolaev's longitudinal induction is introduced:

$$\text{div}\vec{D} = 4\pi\rho + \frac{\varepsilon}{c}\frac{\partial B_{\parallel}}{\partial t} \tag{10}$$

where $B_{\parallel} = \mu_{\parallel} H_{\parallel}$, and scalar magnetic field $H_{\parallel}$ is related to vector potential divergence:

$$H_{\parallel} \sim \text{div}\vec{A} \tag{11}$$

Hypothesis: Parameter $\mu_{\parallel}$ can be interpreted as a coefficient of vacuum "stiffness" relative to longitudinal changes of vector potential (i.e., changes of its divergence), unlike ordinary μ, which characterizes reaction to transverse changes (curl).

Taking divergence from the electric field expression:

$$\vec{E} = -\frac{\mu}{c}\frac{\partial \vec{A}}{\partial t} - \text{grad}\phi \tag{12}$$

and substituting into (10), we obtain the wave equation for scalar potential [6]:

$$\nabla^2\phi - \frac{\varepsilon(\mu - \mu_{\parallel})}{c^2}\frac{\partial^2\phi}{\partial t^2} = -\frac{4\pi\rho}{\varepsilon} \tag{13}$$

Key point: The coefficient at the second time derivative depends on the difference $(\mu - \mu_{\parallel})$. If $\mu_{\parallel} = 0$, we obtain the classical d'Alembert equation. If $\mu_{\parallel} \neq 0$, the wave propagation structure changes.

4 Fermi's Chronometric Field

4.1 Fermi's Variational Principle (1923)

In work [1], Enrico Fermi showed that for correct calculation of electromagnetic mass of an accelerating charge, it is necessary to use **type B variation**. This leads to modification of the integration measure in the action.

Standard interaction action:

$$S_{int} = -\frac{1}{c}\int j^\mu A_\mu \, d^4x \tag{14}$$

Fermi shows that during accelerated motion of charge element de, the proper time element dt is related to laboratory time dt_0 through a geometric factor:

$$dt \longrightarrow dt\left(1 + \frac{\vec{\Gamma}\cdot\vec{r}}{c^2}\right) \tag{15}$$

where $\vec{\Gamma}$ — 4-acceleration, $\vec{r}$ — radius-vector inside the charge from center of mass.

In work [2], it is noted:

> "*Fermi showed an additional integral (equal to -1/3), having a negative sign, which needs to be used to solve the 4/3 problem.*"

4.2 Modification of Integration Measure

In terms of current density $j^\mu = (\rho c, \vec{j})$, this means modification of the volume element in the action:

$$d^4x \longrightarrow d^4x \cdot \mathcal{K}_F, \quad \text{where } \mathcal{K}_F = \left(1 + \frac{\vec{a}\cdot\vec{r}}{c^2}\right) \tag{16}$$

The effective interaction Lagrangian becomes:

$$\mathcal{L}_{int}^{eff} = -\frac{1}{c} j^\mu A_\mu \left(1 + \frac{\vec{a}\cdot\vec{r}}{c^2}\right) \tag{17}$$

4.3 Technical Recipe for Practical Problems

In work [2], the final technical recipe for calculating force between two accelerating charges when integrating only over time is provided:

$$\vec{p}_{12} = \int \vec{F}_{12}^{\text{raw}}(t)\left(1 + \frac{(\vec{\Gamma}_2(t) - \vec{\Gamma}_1(t_{\text{ret}}))\cdot\vec{r}}{c^2}\right) dt \tag{18}$$

where:

- $\vec{\Gamma}_1, \vec{\Gamma}_2$ — accelerations of source and observer,
- $\vec{r}$ — vector between charges,
- t_{ret} — retarded time.

Important: Fermi's correction is a **multiplier in the integration measure**, not a field modification:

> "*Fermi's correction does not change the field, but changes how the force contributes to momentum.*" [2]

5 Connection Between $\mu_\parallel$ and Fermi's Correction

5.1 Formal Derivation

Consider how the modifier $\mathcal{K}_F$ affects field equations. The vector potential can be represented as a sum of transverse and longitudinal parts:

$$\vec{A} = \vec{A}_\perp + \vec{A}_\parallel, \quad \text{where } \mathrm{div}\vec{A}_\perp = 0, \quad \mathrm{rot}\vec{A}_\parallel = 0 \tag{19}$$

The longitudinal part is related to divergence: $\mathrm{div}\vec{A} = \nabla^2\Psi$, where $\vec{A}_\parallel = \mathrm{grad}\Psi$.

In classical electrodynamics (Lorentz gauge), the connection between ϕ and $\mathrm{div}\vec{A}$ is fixed by the condition:

$$\mathrm{div}\vec{A} + \frac{\varepsilon\mu}{c}\frac{\partial\phi}{\partial t} = 0 \tag{20}$$

This condition ensures cancellation of "extra" degrees of freedom and leads to coefficient $\varepsilon\mu/c^2$ in the wave equation.

Fermi's multiplier $\mathcal{K}_F = 1 + \frac{\vec{a}\cdot\vec{r}}{c^2}$ depends on acceleration $\vec{a}$. Charge acceleration is related to field change over time. In linear approximation for a radiating system:

$$\vec{a} \sim \frac{\partial\vec{v}}{\partial t} \sim \frac{\partial}{\partial t}(\text{potentials}) \tag{21}$$

In Fermi's variational principle, the additional term in the action has the form:

$$\delta S_F = -\frac{1}{c}\int j^\mu A_\mu \left(\frac{g_\nu r^\nu}{c^2}\right) d^4x \tag{22}$$

For the longitudinal component of interaction (which is responsible for $\mathrm{div}\vec{A}$ and $\rho\phi$), this term acts as an effective change of connection between current and field.

5.2 Derivation of Expression for $\mu_\parallel$

Compare coefficients at $\frac{\partial^2\phi}{\partial t^2}$ in the wave equation. From work [6] (equation 13):

$$\mathrm{Coeff}_{09} = \frac{\varepsilon}{c^2}(\mu - \mu_\parallel) \tag{23}$$

From Fermi's theory (effective mass $m = U/c^2$ instead of $4/3U/c^2$):

> "*The correction constitutes $-1/3$ of the classical contribution to inertia. The classical contribution is proportional to μ. Consequently, the $\mu_\parallel$ contribution should compensate for this share in the longitudinal channel.*"

In work [2], calculation for a uniformly charged sphere is provided:

$$\int E^{(i)} de = -\frac{4}{3} \cdot 2\frac{u}{c^2}\, a_z \tag{24}$$

where $u = \frac{3}{5}\frac{e^2}{R}$ — field energy of the sphere.
Relation (4) from Fermi's work in modern notation:

$$-\frac{4}{3} \cdot 2\,\frac{u}{c^2}\, a_z + \int \vec{E}^{(i)} \frac{\vec{a} \cdot \vec{R}}{c^2} de + \vec{F} = 0 \tag{25}$$

Formal expression for $\mu_\parallel$:

$$\mu_\parallel = \mu \cdot \lim_{V \to 0} \frac{\int_V \frac{\vec{a} \cdot \vec{r}}{c^2}\, dV}{\int_V dV} \tag{26}$$

For a point charge, this limit requires regularization. In the context of work [6], where vacuum properties are introduced, $\mu_\parallel$ is a fundamental constant arising from space-time geometry in the presence of accelerations.

Using Fermi's 4-notation (g_i — 4-acceleration, r^i — 4-radius):

$$\mu_\parallel = \mu \cdot \kappa \cdot \frac{g_i r^i}{c^2} \tag{27}$$

where κ — coefficient depending on charge distribution geometry (for a sphere, κ leads to elimination of the 4/3 multiplier).

5.3 Interpretation Through Total Current

In work [4], it is shown that Coulomb gauge can be applicable only to the **total vector potential** created by total current:

$$\vec{A}_{\text{total}} = \frac{1}{c} \int \frac{(\vec{j}_{\text{cond}} + \vec{j}_{\text{disp}})}{R} dV \tag{28}$$

Connection hypothesis: Parameter $\mu_\parallel$ can be interpreted as a coefficient necessary for equations for the *conduction* potential to look as if we are working with the *total* potential.

If $\vec{A}_{\text{total}}$ satisfies $\text{div}\vec{A}_{\text{total}} = 0$ (Coulomb gauge for total current), then $\mu_\parallel$ can compensate for the contribution from $\text{div}\vec{A}_{\text{cond}}$.

In the wave equation (13), the multiplier $(\mu - \mu_\parallel)$ reflects the difference between "flat" time (inertial) and "curved" time (accelerated).

6 Applications

6.1 Electromagnetic Impulse of Centrally-Symmetric Explosion

In work [9], a model of centrally-symmetric plasma explosion as a source of electromagnetic impulse is considered. It is shown that classical Liénard-Wiechert potentials give a **positive** impulse sign, whereas Mende's experiment records a **negative** one.

With Fermi's correction, the force becomes:

$$\vec{F}_{\text{eff}} = \int \vec{F}_{\text{LW}}(t) \left(1 + \frac{\vec{a}(t) \cdot \vec{r}}{c^2}\right) dt \tag{29}$$

The additional term $\frac{\vec{a}\cdot\vec{r}}{c^2}$ can change the sign of the resulting impulse. In work [9], it is noted:

> "*Fermi's correction (increase of the gradient integral) can correct the impulse sign without violating conservation laws.*"

6.2 Thrust Calculation in MenDrive Resonator

In work [7], calculation of ponderomotive forces in the waveguide structure of Mende's engine is conducted. Problem geometry:

- Left conductor: $x < -a$
- Vacuum gap: $-a \leq x \leq a$
- Right conductor (ferrite): $x > a$

Maxwell's stress tensor for TE-wave:

$$T_{xx} = \frac{1}{8\pi} \left(\mu_{xx} H_x^2 - \varepsilon_{yy} E_y^2 - \mu_{zz} H_z^2\right) \tag{30}$$

Thrust force:

$$F_x^{\text{thrust}} = T_{xx}\big|_{x=+A^+} - T_{xx}\big|_{x=-A^-} \tag{31}$$

In the current calculation [7], the following are not accounted for:

1. Fermi's chronometric field,
2. Longitudinal magnetic permeability $\mu_\parallel$,
3. Displacement currents of all space.

Proposed modification: Introduce $\mu_\parallel$ for the normal field component at boundaries:

$$T_{xx}^{\text{modified}} = \frac{1}{8\pi} \left(\mu_\parallel H_x^2 - \varepsilon_{yy} E_y^2 - \mu_\perp H_z^2\right) \tag{32}$$

This can create a force imbalance on opposite walls, interpretable as thrust.

7 Unified Scheme

8 Conclusion

This paper has shown that introduction of longitudinal magnetic permeability $\mu_\parallel$ into Maxwell's equations is mathematically equivalent to accounting for Fermi's geometric correction to the integration measure in the action. Main results:

Table 1: Connection Between Concepts

Level	**Concept**	**Mathematical Expression**
Fundamental	Fermi Variation	$dt \to dt(1 + \frac{\vec{a}\cdot\vec{r}}{c^2})$
Field	Longitudinal Permeability	$\mu_{\parallel}$ in $\mathrm{div}\vec{D} = \cdots + \frac{\varepsilon}{c}\frac{\partial B_{\parallel}}{\partial t}$
Potential	Total Current	$\vec{A}_{\text{total}} = \vec{A}_{\text{cond}} + \vec{A}_{\text{disp}}$
Applied	EM Impulse / Thrust	Impulse sign $P \sim \int \vec{E}dt$

1. **The "gauge juggling" problem** is solved through transition to the principle of least action with modified integration measure.

2. **An explicit expression** for $\mu_{\parallel}$ through Fermi's type B variation is derived (equation 26).

3. **Equivalence is shown** between Nikolaev's approach (field modification) and Fermi's (geometric modification).

4. **A modification is proposed** for thrust calculation in MenDrive-type resonators through introduction of $\mu_{\parallel}$ for the normal field component.

Prospects: Experimental verification of modified theory predictions in experiments with centrally-symmetric plasma explosions and precision measurements of ponderomotive forces in asymmetric resonators.

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